An index is a position of an individual character or element in a list, tuple, or string. The index value always starts at zero and ends at one less than the number of items.

Fig: Index position

Negative indexes enable users to index a list, tuple, or other indexable containers from the end of the container, rather than the start.

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What Is Slicing?

Slicing is the extraction of a part of a string, list, or tuple. It enables users to access the specific range of elements by mentioning their indices.

Syntax: Object [start:stop:step]

- "Start" specifies the starting index of a slice
- "Stop" specifies the ending element of a slice
- You can use one of these if you want to skip certain items

Example:

Fig: Python slicing

Note: The search will start at index one (included) and ends at index six (not included).

The following image shows some examples of slicing in a string and tuple:

Fig: Additional slicing examples

Slice with Negative Indices

Negative slicing begins at the end of the list.

Example:

Fig: Slicing using negative indices

Specify Step of the Slicing

You can follow a specific step in order to skip certain items.

Example:

Fig: Stepped slicing

In the above example, the string 'SIMPLILEARN' is sliced from indexes one through seven. However, because the step size was set as number two, you will receive every second character in the output starting from the first index.

Reversing Elements of Data Structure

We can use a negative step to reverse the elements of a data structure.

Example:

Fig: Reversing elements using negative step

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Slice() Function in Python

The **slice()** method extracts a section of data and returns it as new data, without modifying it. This means users can take a specific range of elements without changing it.

Syntax: slice(start,stop,step)

Example:

Fig: slice() function

Insert and Delete Multiple Items Using Slicing

1. Slice insertion in Python

Users can insert items into a list without replacing other elements.

Example:

Fig: Inserting elements using Python slicing

2. Slice deletion in Python

Users can delete multiple items out of the data structure by using a **del** statement.

Example:

Fig: Delete list items using Python slicing

Note: Tuple object does not support item deletion.

Conclusion

In this article, we discussed two key concepts: Python slicing and indexing. We looked at how slicing, negative slicing, and step-indexing works through various examples. Both concepts are essential for understanding Python.

If you have any questions, please leave them in the comments section, and we'll have our experts answer them for you.

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